

Title:

Collections & Exception Handling

Objective

To demonstrate use of collections and exception handling.

Concepts Used

- ArrayList
- Try-Catch
- Exception handling

Code

```
import java.util.*;

public class CollectionExceptionDemo {
    public static void main(String[] args) {

        ArrayList<String> students = new ArrayList<>();
        Scanner sc = new Scanner(System.in);

        try {
            // Adding elements
            students.add("Sushanth");
            students.add("Rahul");
```

```
System.out.println("Student List: " + students);

// Taking input
System.out.print("Enter index to view: ");
int index = sc.nextInt();

// This may cause exception
System.out.println("Student at index: " + students.get(index));

} catch (IndexOutOfBoundsException e) {
    System.out.println("Error: Invalid index!");
} catch (Exception e) {
    System.out.println("General Error occurred");
} finally {
    System.out.println("Program executed successfully.");
}
}
}
```

Output

Case 1 - invalid

```
C:\Users\susha\OneDrive - Amrita Vishwa Vidyapeetham\Desktop\New folder>java CollectionExceptionDemo
Student List: [Sushanth, Rahul]
Enter index to view: 5
Error: Invalid index!
Program executed successfully.
```

Case 2 - valid

```
C:\Users\susha\OneDrive - Amrita Vishwa Vidyapeetham\Desktop\New folder>java CollectionExceptionDemo
Student List: [Sushanth, Rahul]
Enter index to view: 1
Student at index: Rahul
Program executed successfully.
```